Load and Cache a CHELSA-TraCE21k Raster

Description

Loads a cropped CHELSA-TraCE21k raster for a given bioclim variable and age slice. If a cached '.tif' already exists in 'dir_cache' it is returned immediately without re-downloading. Absolute-temperature variables ('bio1', 'bio6') are corrected from Kelvin to degrees Celsius (subtract 273.15) before the raster is written to cache.

Usage

```
load_chelsa_raster(

  abiotic_variable_name = NULL,

  age = NULL,

  x_lim = NULL,

  y_lim = NULL,

  dir_cache = NULL

)
```

Arguments

`abiotic_variable_name`	Character scalar. Project-level bioclim variable name, e.g. '"bio1"', '"bio4"', '"bio12"'. Single-digit numbers are zero-padded internally to match CHELSA file names ('"bio1"' becomes '"bio01"' in the URL).
`age`	Numeric or integer scalar. Age in years BP used to encode the CHELSA-TraCE21k time step (e.g. '1000' encodes as time step '"-010"').
`x_lim`	Numeric vector of length 2. Longitude extent 'c(min, max)' for cropping the downloaded raster.
`y_lim`	Numeric vector of length 2. Latitude extent 'c(min, max)' for cropping the downloaded raster.
`dir_cache`	Character scalar. Path to the directory where the cropped raster is cached as 'abiotic_variable_name_age_xmin_xmax_ymin_ymax.tif'. The extent values are rounded to 2 decimal places so the cache is shared across calls with identical extents. The directory must already exist before calling this function.

Details

CHELSA-TraCE21k time steps: 'age = 0' (present) maps to the special step '"0000"'; all other ages use 'sprintf("- '"-001"' and 'age = 1000' gives '"-010"'.

The remote raster is accessed via GDAL '/vsicurl/', so an internet connection is required the first time each '(abiotic_variable_name, age, x_lim, y_lim)' combination is requested. Subsequent calls load from the cached '.tif' and need no connection. The geographic extent (rounded to 2 decimal places) is embedded in the cache filename, so rasters with different extents are always stored separately.

Kelvin correction: 'bio1' (Mean Annual Temperature) and 'bio6' (Min Temperature of Coldest Month) are absolute temperatures stored in Kelvin; 273.15 is subtracted before caching. Range and seasonality variables ('bio4', 'bio7') are differences or standard deviations — no offset needed.

Value

A 'terra::SpatRaster' cropped to 'x_lim' / 'y_lim', with corrected units: degrees Celsius for 'bio1' and 'bio6'; original CHELSA units for all other variables.

See Also

[interpolate_mev_to_grid()], [interpolate_st_mev_to_grid()], [project_coords_to_metric()]